Q6. Box Plot and Outlier Identification

Original Aim: To examine the spread of an Iris numerical attribute and identify possible outliers. Draw a box plot for SepalWidthCm and calculate Q1, Q3 and the interquartile range (IQR). Use the 1.5 × IQR rule to identify any possible outlier values and report the result.

Aim

Examine SepalWidthCm with a box plot, calculate quartiles/IQR and identify values outside 1.5 times IQR fences.

Requirements

Python, Pandas, scikit-learn and Matplotlib. Reference: [D6].

Essential theory

Q1 and Q3 are the 25th and 75th percentiles. IQR=Q3-Q1 describes the middle 50% spread. Lower fence=Q1-1.5*IQR; upper fence=Q3+1.5*IQR. Values strictly outside these fences are possible statistical outliers. Standard box-plot whiskers reach the most extreme observed values still inside the fences, not necessarily the fences themselves.

Step-by-step procedure

1. Select SepalWidthCm from the complete dataset.
2. Use quantile with linear interpolation to calculate Q1/Q3; derive IQR and fences.
3. Select rows satisfying x<lower or x>upper, preserving original row indices and species.
4. Draw a horizontal box plot with whis=1.5 and mark the calculated fences.
5. Report the count and exact flagged values without deleting them.

Complete Python program

```
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
features = [
    "SepalLengthCm", "SepalWidthCm",
    "PetalLengthCm", "PetalWidthCm"
]
species = ["Iris-" + name for name in iris.target_names]
df = pd.DataFrame(iris.data, columns=features)
df["Species"] = pd.Categorical.from_codes(
    iris.target, categories=species
)

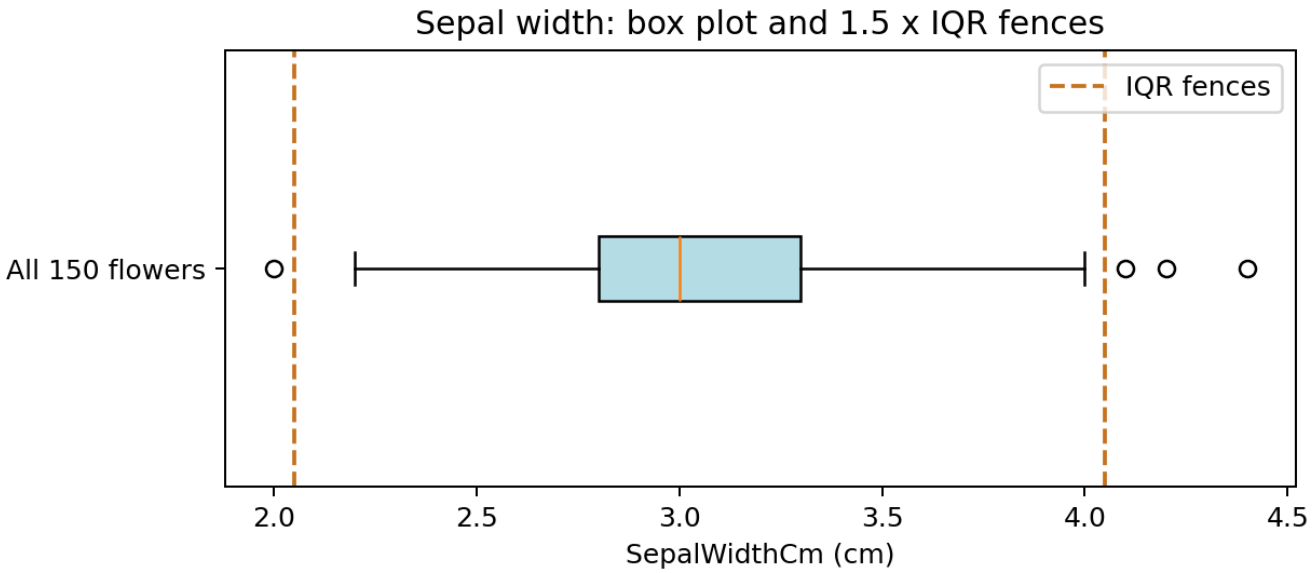
x = df["SepalWidthCm"]
q1 = x.quantile(0.25, interpolation="linear")
q3 = x.quantile(0.75, interpolation="linear")
iqr = q3 - q1
lower = q1 - 1.5 * iqr
upper = q3 + 1.5 * iqr
outliers = df.loc[(x < lower) | (x > upper),
                  ["SepalWidthCm", "Species"]]
print(f"Q1={q1:.4f}; Q3={q3:.4f}; IQR={iqr:.4f}")
print(f"Lower fence={lower:.4f}; Upper fence={upper:.4f}")
print("Possible outlier count:", len(outliers))
print(outliers.to_string())
fig, ax = plt.subplots(figsize=(7, 3.2))
ax.boxplot(x, vert=False, whis=1.5, patch_artist=True,
           boxprops={"facecolor": "#b4dce4"})
ax.axvline(lower, color="#c57221", linestyle="--", label="IQR fences")
ax.axvline(upper, color="#c57221", linestyle="--")
ax.set_title("Sepal width: box plot and 1.5 x IQR fences")
ax.set_xlabel("SepalWidthCm (cm)")
ax.set_yticks([1], ["All 150 flowers"])
ax.legend(loc="upper right")
fig.tight_layout()
fig.savefig("q06_boxplot.png", dpi=180)
plt.show()
```

Code explanation

- `quantile(..., interpolation="linear")` interpolates between ordered values at fractional index positions, following the selected Pandas convention.
- `(x < lower) | (x > upper)` combines two Series conditions with bitwise OR; parentheses are required. Plain Python `or` does not combine Series elementwise.
- `df.loc[mask, ...]` returns the exact matching records, so index numbers are preserved.
- `boxplot(..., whis=1.5)` uses the same IQR rule; circles represent flagged observations, while dashed lines show fences.

Verified output and figure

```
Q1=2.8000; Q3=3.3000; IQR=0.5000
Lower fence=2.0500; Upper fence=4.0500
Possible outlier count: 4
    SepalWidthCm          Species
15           4.4      Iris-setosa
32           4.1      Iris-setosa
33           4.2      Iris-setosa
60           2.0  Iris-versicolor
```



Observation

Q1=2.8 and Q3=3.3 cm give IQR=0.5 cm. The fences are 2.05 and 4.05 cm. Row 60 (2.0 cm) lies below the lower fence; rows 15, 32 and 33 (4.4, 4.1 and 4.2 cm) lie above the upper fence. The whiskers extend to observed non-outliers 2.2 and 4.0 cm. These are pooled-data flags, not a finding that the flower measurements are wrong.

Result

Four possible SepalWidthCm outlier observations were identified consistently by the calculation and box plot. No rows were removed; investigate data quality, units and species context before considering exclusions.

Common errors and troubleshooting

- Quartiles differ from another package: compare interpolation conventions and dataset versions.
- Four numeric values but fewer visible markers: overlapping equal-valued observations can share a plotting position; count rows from the mask.
- Every high value is deleted: flagging is not confirmation of error; keep the records unless an explicit justified rule is assigned.

Viva questions and answers

- What is IQR? The difference Q3-Q1, describing the central 50% range.
- Does a box-plot whisker end at the theoretical fence? Usually it ends at the extreme observed non-outlier.
- Are values exactly on a fence flagged here? No; the code uses strict less-than and greater-than.
- Is a statistical outlier necessarily a data-entry mistake? No; it can be a valid unusual observation.